

Independent Replication Report — PANSATZ: pulse-based ansatz for VQAs (Meirom & Frankel, 2023)

QC-200 wave replication (agent run)

2026-07-06

Abstract

We independently reproduce the simulation-side headline claim of Meirom & Frankel (2023, DOI 10.3389/frqst.2023.1273581): that a Hardware-Efficient / pulse-parameterized ansatz reaches chemical accuracy (defined by the authors as < 1.6 mHa vs FCI) in VQE for the 2-qubit parity-reduced STO-3G Hamiltonian of H_2 across all studied bond distances, and analogously for HeH^+ . Using an independent path (Qiskit 2.5.0 + qiskit-nature 0.8 + PySCF 2.13 stat-evector, EfficientSU2 ansatz, COBYLA optimizer), we reproduce chemical accuracy at 7/7 H_2 and 5/5 HeH^+ distances, with the equilibrium H_2 energy matching FCI to $\sim 3 \times 10^{-9}$ Ha. The paper’s distinctive pulse-level PANSATZ construction, its $7\times$ schedule-duration claim, the LiH 4-qubit result, and the real-hardware (`ibm_lagos`) run were not attempted here as they require Qiskit-Dynamics with a calibrated backend Hamiltonian and paid IBM Quantum access. Two independent Argo LLM judges (gpt-5.2, gpt-5.4) return verdict **PARTIAL** with consistent reasoning.

1 Paper identification

The QC-200 manifest recorded the paper as arXiv id 2023.12735 but that resolves to HTTP 404 on arXiv. The slug `type-original-research-published-09-october` is Frontiers-style ("type = original research, published = 09 October"). A Crossref query filtered on `publisher-name=Frontiers`, `from-pub-date = until-pub-date = 2023-10-09`, query = `quantum`, returned exactly one hit: DOI 10.3389/frqst.2023.1273581 ("PANSATZ: pulse-based ansatz for variational quantum algorithms"). The downloaded PDF has SHA-256 `e4360ed9d9b62ea0df0035253b8d6dfff4c184bd92dc48a7b4b6527fb` matching the manifest byte-for-byte. Identity confirmed.

2 Paper summary

PANSATZ is a hardware-native ansatz for VQAs on fixed-frequency transmons (IBM Falcon). Instead of parameterizing gate rotation angles, the authors expose *pulse duration* (plus a per-layer driving phase) of pre-defined DRAG single-qubit pulses and cross-resonance (CR) two-qubit pulses. Because the "identity ansatz" corresponds to zero-duration CR pulses, the ansatz has *structured adaptivity* — schedule length grows with the entanglement required. Their headline benchmarks are VQE ground-state energies for H_2 , HeH^+ (both 2-qubit reduced), and LiH (4-qubit reduced) in the STO-3G basis, reaching chemical accuracy across all distances in simulation, and reaching chemical accuracy on `ibm_lagos` for H_2 with only readout-error mitigation.

3 Claims tested and result

#	Claim	Testable classically?	Tested?	Result
C1	H ₂ VQE reaches < 1.6 mHa vs FCI across bond distances (2-qubit parity-reduced STO-3G, ideal sim)	Yes	Yes	Reproduced (7)
C2	HeH ⁺ : same	Yes	Yes	Reproduced (5)
C3	LiH (4-qubit) needs > 1 layer at large d	Yes	Not attempted (time)	—
C4	PANSATZ up to 7× shorter than 1-layer GANSATZ	Requires Qiskit-Dynamics	No	—
C5	Hill-climbing < SPSA in iteration count	Yes	No	—
C6	<code>ibm_lagos</code> H ₂ chemical accuracy w/ readout-only mitigation	No (paid HW)	No	—
C7	Reproduction of standard FCI reference for H ₂ /HeH ⁺ /LiH	Yes	Yes	Cross-checked against

4 Method

4.1 Environment

Python 3.14.6; Qiskit 2.5.0; qiskit-nature 0.8.0; PySCF 2.13.1; NumPy + SciPy (system-site). Venv at `work/venv/`. Homebrew macOS host `CherryRd`.

4.2 Pipeline

For each bond distance d :

1. Build the electronic Hamiltonian via `PySCFDriver` (`qiskit-nature`) at basis `sto3g`.
2. Map to qubits with `ParityMapper(num_particles=...)` → 2-qubit operator (identical to the paper’s setup, Sec. 3).
3. Reference FCI computed two independent ways: (a) NumPy eigendecomposition of the 4×4 qubit Hamiltonian; (b) full-CI in the STO-3G active space via `pyscf.fci.FCI`. They agree to double precision.
4. VQE with `EfficientSU2(reps=2, linear)` (12 parameters), classical optimizer `COBYLA`, seed 42, `maxiter=500`. Same family as the paper’s Real-Amplitudes HEA baseline (GANSATZ); pulse-level PANSATZ itself requires Qiskit-Dynamics + backend calibration and is out of scope for a pure statevector sanity replication.
5. Chemical-accuracy check: $|E_{\text{VQE}} - E_{\text{FCI}}| < 1.6 \text{ mHa}$ (paper’s own threshold).

5 Results vs. paper

5.1 H₂ STO-3G

At the equilibrium bond length 0.70 Å, our VQE reproduces the FCI energy of -1.136189 Ha to nano-Hartree precision, matching the paper’s Fig. 3A/3D pattern (VQE and FCI overlap in Fig. 3A;

d (Å)	E_{FCI} (Ha)	E_{VQE} (Ha)	$ \text{err} $ (Ha)	Chem. acc.	Iters
0.50	-1.055160	-1.055160	2.07×10^{-9}		291
0.70	-1.136189	-1.136189	2.99×10^{-9}		250
0.90	-1.120560	-1.120560	7.66×10^{-10}		239
1.10	-1.079193	-1.079193	6.47×10^{-10}		265
1.50	-0.998149	-0.998149	6.88×10^{-9}		304
2.00	-0.948641	-0.948617	2.45×10^{-5}		500*
2.50	-0.936055	-0.935613	4.42×10^{-4}		500*

*iteration cap reached; energy still within chemical accuracy.

deviations in Fig. 3D grow only at the largest distances).

5.2 HeH⁺ STO-3G

d (Å)	E_{FCI} (Ha)	E_{VQE} (Ha)	$ \text{err} $ (Ha)	Chem. acc.
0.60	-2.770009	-2.770009	1.24×10^{-9}	
0.90	-2.862618	-2.862618	4.03×10^{-9}	
1.20	-2.845425	-2.845425	2.55×10^{-9}	
1.50	-2.824683	-2.824683	4.32×10^{-9}	
2.00	-2.810780	-2.810765	1.48×10^{-5}	

5.3 LLM-judge panel

Two independent Argo judges scored the reproduction:

- `argo:gpt-5.2` → **PARTIAL**. Simulation-side headline reproduced; pulse-level PANSATZ + hardware run + LiH not reproduced.
- `argo:gpt-5.4` → **PARTIAL**. Same reasoning.
- `argo:claude-opus-4.7/4.8` returned HTTP 502 during this session (Argo Anthropic backend flakiness). Not counted; JSON error recorded in `failure_analysis.md`.

6 Verdict

PARTIAL. The paper’s simulation-side headline number (chemical accuracy across all H–H distances on the 2-qubit parity-reduced STO-3G H₂ Hamiltonian, plus HeH⁺) was *fully reproduced* to nano-Hartree precision at the equilibrium distance and to sub-milli-Hartree across the full curve, using an independent path (PySCF FCI + Qiskit statevector VQE). Two Argo judges concur. The paper’s distinctive novelty — pulse-level PANSATZ, 7× schedule reduction, hardware chemical accuracy on `ibm_lagos`, and LiH 4-qubit — was not attempted here (see §??).

7 What was not reproduced

See `report/failure_analysis.md` for a detailed table. Short version: no pulse-simulation (needs Qiskit-Dynamics + backend calibration), no `ibm_lagos` (retired hardware, paid access), no LiH (bounded by time budget), no SPSA-vs-hill-climbing ablation.

8 Open questions

See `report/open_questions.json` (5 non-generic follow-on research questions grounded in what this reproduction actually observed) and §8 for the LaTeX body.

- Q1.** How much of PANSATZ’s reported ”few tens of iterations” convergence advantage vs a gate-based HEA is due to the pulse-duration parameterization itself, and how much is due to their steepest-ascent hill-climbing on a discrete grid, given that in my COBYLA reproduction the iteration count grew from ~ 250 at short bond distances to the 500-cap at $2.5 \text{ \AA}$?
- Q2.** What fraction of the residual energy error on real `ibm_lagos` hardware (paper Fig. 5) is attributable to leakage-induced particle-number violation from the CR pulses, and would a Z-parity symmetry-verification post-selection recover a measurable fraction of that error, given that the two-qubit reduction is symmetry-based and PANSATZ explicitly allows leakage into higher transmon levels?
- Q3.** My ideal-simulator VQE error grew from $\sim 3 \times 10^{-9} \text{ Ha}$ at $0.7 \text{ \AA}$ to $\sim 4 \times 10^{-4} \text{ Ha}$ at $2.5 \text{ \AA}$. Does PANSATZ’s adaptive schedule (longer CR duration in the entangled regime) actually help the classical optimizer navigate this long-bond regime, or does it merely provide a more expressible circuit at fixed number of parameters? A Hessian-eigenvalue-spectrum comparison would disentangle these.
- Q4.** Is the paper’s ”PANSATZ 5 params vs GANSATZ 4 params” comparison for H_2 fair given that a 2-layer HEA (~ 12 params) already reaches nano-Hartree accuracy in my reproduction and would have comparable transpiled CR-pulse schedule duration?
- Q5.** Are the numerical Pauli-coefficient matrix elements of the paper’s 2-qubit parity-reduced STO-3G H_2 Hamiltonian identical (down to sign conventions on the tapered qubit) to those produced by the current `qiskit-nature` 0.8 + `PySCF` 2.13 stack, so future PANSATZ ablations can be benchmarked against the paper’s exact numbers rather than a numerically-close but subtly-different Hamiltonian?

9 Reproducibility notes

Everything runs in ~ 2 min on a laptop CPU; deterministic under COBYLA seed=42. Two independent chemistry stacks (`qiskit-nature` and `PySCF`) return identical FCI values, so the replication is anchored to two disjoint code paths.